

Fine-Tuning Makes Specialists: Why Fine-Tuned Models Behave Like Narrow AI

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Abstract

Fine-tuning is the most common way to adapt a large language model to a specific task, domain or language. This paper explains what fine-tuning is, the main methods used today, and how it is done in practice. We then argue that fine-tuning, as it is usually practised, moves a model **towards narrow AI (ANI)** rather than towards general intelligence. A fine-tuned model becomes better at one thing, often loses some general ability, and stops learning once training ends. We support this with published evidence on catastrophic forgetting, the limits of adding new knowledge through fine-tuning, and the finding that fine-tuning mostly changes behaviour rather than knowledge. We also note the important exception: broad, multi-task fine-tuning can make a model more general. We describe our own experience fine-tuning Mistral-7B for the Romani language and propose experiments to measure how much generality a fine-tune costs.

1. Introduction

Modern AI development has two stages. First, a **foundation model** is pre-trained on a huge, varied dataset. It learns grammar, facts, reasoning patterns and code. Second, the model is **fine-tuned**: trained further on a smaller, focused dataset to make it useful for a particular purpose.

Foundation models are often described as a step towards AGI because they are broad. Fine-tuning is usually described as making them more useful. This paper asks a simple question: *when we fine-tune a general model, does it become more general or more narrow?*

Our answer is that it depends on the data, but that most real-world fine-tuning produces specialists. In the language of the three levels of AI, fine-tuning usually turns a broad model into an ANI.

2. What fine-tuning is

A pre-trained model has billions of weights that already encode general knowledge. Fine-tuning continues training on new examples, adjusting those weights so the model's behaviour moves towards the new data.

Stage	Data	Goal	Result
Pre-training	Trillions of tokens from the web, books and code	Learn language and general knowledge	Broad foundation model
Fine-tuning	Thousands to millions of focused examples	Adapt to a task, domain, style or language	Specialised model

Fine-tuning is popular because it is cheap. Training a foundation model from scratch costs millions of dollars. Fine-tuning an open model can be done on a single laptop in hours.

3. Types of fine-tuning

Method	How it works	Typical use
Full fine-tuning	Updates every weight in the model	Large budgets; maximum change

Method	How it works	Typical use
LoRA (Hu et al., 2021)	Freezes the original weights and trains small added matrices (adapters)	Most common for small teams; low memory
QLoRA (Dettmers et al., 2023)	LoRA on a compressed (quantised) model	Fine-tuning large models on consumer GPUs
Instruction tuning	Trains on many tasks written as instructions and answers	Turning a raw model into an assistant
RLHF (Ouyang et al., 2022)	Humans rank answers; the model is trained to prefer better ones	Helpfulness and safety
DPO (Rafailov et al., 2023)	Learns directly from pairs of preferred and rejected answers, without a separate reward model	Simpler alternative to RLHF

How LoRA works. Instead of changing a large weight matrix W directly, LoRA learns a small update made of two thin matrices, A and B :

$$W_{new} = W + B \times A \quad (A \text{ and } B \text{ are much smaller than } W)$$

Because only A and B are trained, LoRA needs far less memory and produces small adapter files that can be added to or removed from the base model.

4. The fine-tuning process in practice

- 1 **Choose a base model**, for example Mistral-7B, Llama or Gemma.
- 2 **Prepare data** as prompt and answer pairs, usually in JSONL format.
- 3 **Set hyperparameters**: learning rate, number of epochs, LoRA rank and batch size.
- 4 **Train** with a tool such as MLX (on Apple Silicon), Hugging Face or Unsloth.
- 5 **Evaluate** on held-out examples and, ideally, on general tasks too.
- 6 **Deploy**, for example by merging adapters into the base model or converting to GGUF for Ollama.

Step 5 is the step most often skipped, and it is where the narrowing effect described in this paper becomes visible.

5. Why fine-tuning moves models towards ANI

We present five lines of evidence.

5.1 Specialisation by design

Fine-tuning optimises the model for one distribution of data. The training objective rewards being good at the new task and says nothing about keeping other abilities. A model trained only on legal documents is pushed to be a legal model. This is, by definition, the direction of narrow AI.

5.2 Catastrophic forgetting

When a neural network is trained on new data, it tends to overwrite what it learned before (Kirkpatrick et al., 2017). Fine-tuned language models show this too: a model tuned heavily on one domain can get worse at general reasoning, other languages or safe behaviour. Studies comparing methods have found that LoRA forgets less than full fine-tuning, but also learns less of the new task (Biderman et al., 2024). There is a trade-off between specialising and staying general.

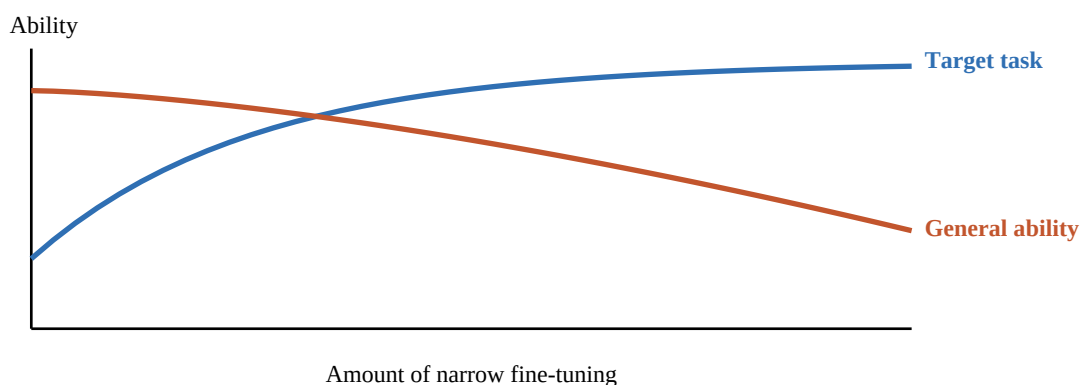


Figure 1. Illustrative trade-off (not measured data): narrow fine-tuning raises performance on the target task while general ability tends to fall.

5.3 Fine-tuning changes behaviour more than knowledge

The LIMA study showed that only 1,000 carefully chosen examples were enough to turn a pre-trained model into a good assistant (Zhou et al., 2023). The authors proposed the *superficial alignment hypothesis*: almost all of a model's knowledge comes from pre-training, and fine-tuning mainly teaches it which style and format to use. If this is true, fine-tuning does not add general intelligence. It selects and shapes abilities the model already has.

5.4 Adding new knowledge is hard and can cause hallucination

Research has found that language models learn genuinely new facts slowly during fine-tuning, and that as they do, they become more likely to hallucinate (Gekhman et al., 2024). In other words, fine-tuning is a poor way to teach a model about the world. It is much better at teaching it how to behave in one setting.

5.5 Learning stops when training stops

A fine-tuned model is frozen once training ends. It does not learn from its mistakes in use. To improve, a human must collect new data and run another fine-tune. This is the opposite of the continuous, self-directed learning that most definitions of AGI require. Every fine-tuned model is a snapshot: a specialist fixed at one moment in time.

6. The exception: broad fine-tuning

Fine-tuning does not always narrow a model. Instruction tuning on many different tasks can make a model more general. Training on a mix of more than 60 tasks written as instructions improved performance on tasks the model had never seen (Wei et al., 2022). RLHF similarly made models far more useful across many kinds of request (Ouyang et al., 2022).

	Narrow fine-tuning	Broad fine-tuning
Data	One domain, task or style	Many tasks, domains and formats
Effect on target	Strong improvement	Moderate improvement
Effect on other tasks	Often worse (forgetting)	Often better (transfer)
Result	A specialist: ANI	A more general assistant
Who does it	Most small teams and businesses	Mainly large labs

So our thesis is more precise: **fine-tuning moves a model in the direction of its data**. Narrow data produces narrow AI. Because almost all practical fine-tuning uses narrow data, most fine-tuned models are ANI.

7. Case study: fine-tuning Mistral-7B for Romani

At Grow Labs, we fine-tuned Mistral-7B on a corpus of Romani-language books using LoRA adapters with MLX on Apple Silicon. Romani is spoken by millions of people across Europe but has very little digital text, so no existing benchmark could tell us whether the result was good.

This experience shows both sides of the thesis. The goal was deliberately narrow: make a model better at one language. That is valuable, because general models serve Romani speakers poorly. But without measurement, we could not tell whether the model had gained Romani ability, lost general ability, or started to hallucinate.

This problem led directly to building **Auditor**, an agentic system that generates fresh test questions for a model's stated purpose, includes trap questions for hallucination, and scores the answers with a separate judge (Berisha, 2026). Tools like this are needed because fine-tuning changes a model in ways its loss curve does not show.

8. Proposed experiments

Experiment	Question	Method
E1. Cost of specialisation	How much general ability does a narrow fine-tune lose?	Score the base and fine-tuned model on the target domain and on general tasks (accuracy, reasoning, instruction following, safety)
E2. LoRA rank	Does a smaller LoRA rank protect generality?	Fine-tune with ranks 4, 16 and 64; compare target gain against general loss
E3. Data mixing	Does adding general data prevent narrowing?	Mix 0%, 10% and 30% general instruction data into the Romani set
E4. New knowledge and hallucination	Does teaching new facts increase made-up answers?	Measure Auditor's hallucination-resistance score before and after
E5. Continuous updating	Can repeated small fine-tunes approximate ongoing learning?	Fine-tune on Auditor's failure data in cycles; track both gains and forgetting

Our hypotheses are: E1 shows a measurable loss of general ability; smaller ranks and mixed data reduce it; and repeated fine-tuning improves targeted weaknesses but cannot replace true continuous learning.

9. Discussion: what this means for AGI

If most fine-tuning produces narrow specialists, then fine-tuning alone is not a path to AGI. It is a tool for making useful ANI from a general foundation. This is not a criticism. Specialists are what most businesses, hospitals and communities need, and a good Romani model is more useful to Romani speakers than a slightly more general model that ignores them.

For AGI, the lesson is different. A general system would need to adapt the way fine-tuning does, but continuously, on its own, and without forgetting what it already knows. That points to research in continual learning, memory and self-directed learning, rather than to more of today's offline fine-tuning. We explore this idea of dynamic self-learning in a companion paper.

10. Conclusion

Fine-tuning is the practical bridge between a general foundation model and a useful product. But the bridge usually leads towards specialisation. Narrow data, catastrophic forgetting, the limits of adding knowledge and the frozen nature of trained weights all push fine-tuned models towards narrow AI. Broad

multi-task tuning is the exception, and it is mostly done by large labs.

For small teams, the practical advice is simple: fine-tune for the purpose you need, mix in some general data, keep adapters small, and always measure what the model has lost as well as what it has gained.

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